Matrix Decompositions: A Field Guide

This guide collects the matrix factorisations that come up most often in numerical linear algebra and machine learning: LU, QR, the eigendecomposition and the singular value decomposition, with what each one is good for.

1. LU decomposition
A square matrix A can often be written as A = LU, with L lower triangular and U upper triangular. It is Gaussian elimination written down, and it solves many systems with the same A quickly.

2. QR decomposition
Any matrix can be written as A = QR, with Q having orthonormal columns and R upper triangular. Gram-Schmidt or Householder reflections compute it; it is the stable way to solve least-squares problems.

3. Eigendecomposition
A diagonalisable square matrix is A = V Lambda V^-1, with the eigenvalues on the diagonal of Lambda.

4. Singular value decomposition
Definition. Every real m x n matrix A can be written as A = U Sigma V^T, where U (m x m) and V (n x n) are orthogonal and Sigma is an m x n diagonal matrix with non-negative entries sigma_1 >= sigma_2 >= ... >= 0, the singular values of A. The columns of U and V are the left and right singular vectors.

5. Applications

Keeping the k largest singular values gives the best rank-k approximation of A (Eckart-Young). The same idea underlies principal component analysis, the pseudo-inverse and simple image compression.